



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 28

Level 2 — Intermediate · Logical Reasoning I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Logical Reasoning I — Series, Coding & Directions
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	28 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 28 upgrades reasoning to Level 2: series with GROWING gaps and alternating rules, backward letter codes and word reversal, both-ends ranking (8th from left AND right), compound direction turns, the mirror-clock classic, and two-rule number machines. These mirror the mid-difficulty band of the IMO Logical Reasoning section.

Learning Objectives

- Crack series with growing differences and alternating two-step rules.
- Decode backward shifts, reversals and letter-value sums.
- Solve both-ends ranking and multi-person ordering puzzles.
- Track direction through multiple turns and use the 6-8-10 triangle.
- Reason about mirror clocks and rotated compass directions.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

Q1. Find the next number: **3, 4, 6, 9, 13, ____**

- (A) 16 (B) 17
(C) 18 (D) 19

Q2. Find the next number: **2, 6, 4, 8, 6, 10, ____**

- (A) 8 (B) 12
(C) 14 (D) 7

Q3. If **PEN** is coded as **ODM** (each letter one step BACK), how is **BOOK** coded?

- (A) ANNJ (B) CPPL
(C) ANOJ (D) AMNJ

Q4. In a code, **TIGER** is written as **REGIT**. How is **HORSE** written?

- (A) ESROH (B) ESORH
(C) EHSRO (D) ESHRO

Q5. If A=1, B=2, C=3 ..., what is the CODE for the word **BED** as a sum?

- (A) 10 (B) 11
(C) 12 (D) 13

Q6. In a class of 30 students, Tanvi's rank is 11th from the TOP. What is her rank from the BOTTOM?

- (A) 19th (B) 20th
(C) 21st (D) 18th

Q7. Five friends finished a race. Mira beat Nia. Nia beat Ojas. Pari beat Mira. Ojas beat Qadir. Who finished LAST?

- (A) Nia (B) Ojas
(C) Qadir (D) Mira

Q8. Raj walks 6 m North, turns RIGHT and walks 8 m. In which direction is he walking now, and how far is he from the start (straight line)?

- (A) East; 10 m (B) West; 10 m
(C) East; 14 m (D) South; 10 m

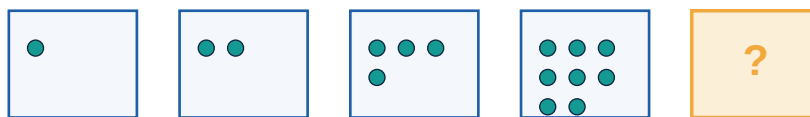
Q9. A girl faces West. She turns RIGHT, then RIGHT again, then LEFT. Which direction does she face now?

- (A) North (B) South
(C) East (D) West

Q10. Find the missing number in the series: **1, 2, 4, 7, 11, ___, 22**

- (A) 15 (B) 16
(C) 17 (D) 18

Q11. Which figure comes next? (Count the dots.)



- (A) 10 dots (B) 12 dots
 (C) 14 dots (D) 16 dots

Q12. If **yesterday's tomorrow** is Wednesday, what day is TODAY?

- (A) Tuesday (B) Wednesday
 (C) Thursday (D) Monday

Q13. Complete: **AZ, BY, CX, __**

- (A) DV (B) DW
 (C) EW (D) DX

Q14. In a row of girls, Veda is 8th from the left AND 8th from the right. How many girls are in the row?

- (A) 14 (B) 15
 (C) 16 (D) 17

Q15. If South-East becomes NORTH, North-East becomes WEST (everything rotates the same way), what does EAST become?

- (A) North-West (B) South-West
 (C) North-East (D) South

Q16. Find the odd one out: **16, 25, 36, 48, 49**

- (A) 25 (B) 36
 (C) 48 (D) 49

Q17. A clock is kept in front of a mirror. The clock shows 3:00. What time does the MIRROR image appear to show?

- (A) 3:00 (B) 9:00
 (C) 6:00 (D) 12:00

Q18. Aman is twice as old as Bela. Bela is 6 years old. How old will Aman be after 3 years?

- (A) 12 (B) 15
 (C) 18 (D) 9

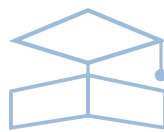
Q19. Which number replaces the question mark? **4 → 17, 6 → 25, 8 → ?** (Hint: $\times 4$, then $+ 1$)

- (A) 31 (B) 32
 (C) 33 (D) 35



Q20. Three boxes: the RED box is heavier than the BLUE box. The GREEN box is lighter than the BLUE box. Which statement **MUST** be true?

- (A) Green is the heaviest (B) Red is the heaviest
(C) Blue is the heaviest (D) All weigh the same



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	A	A	A	B	B	C	A	A	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	D	B	B	B	A	C	B	B	C	B

Step-by-Step Solutions

Q1. Correct answer: (C)

The gaps grow by 1 each time: +1, +2, +3, +4, so next is +5: $13 + 5 = 18$.

Q2. Correct answer: (A)

The rule alternates: +4, -2, +4, -2, +4 ... After 10, apply -2: **8**.

Q3. Correct answer: (A)

One step back each: B→A, O→N, O→N, K→J. BOOK → **ANNJ**.

Q4. Correct answer: (A)

The word is reversed: HORSE → **ESROH**.

Q5. Correct answer: (B)

B=2, E=5, D=4. Sum = $2 + 5 + 4 = 11$.

Q6. Correct answer: (B)

$(30 - 11) + 1 = 20$ th from the bottom.

Q7. Correct answer: (C)

Order: Pari > Mira > Nia > Ojas > Qadir. The last is **Qadir**.

Q8. Correct answer: (A)

From North, a right turn faces **East**. The 6 m and 8 m legs form a right angle, and 6-8-10 is a right triangle, so he is **10 m** from the start.

Q9. Correct answer: (A)

West → right → North → right → East → left → **North**.

Q10. Correct answer: (B)

Gaps: +1, +2, +3, +4, +5, +6. $11 + 5 = 16$, and $16 + 6 = 22$ ✓.

Q11. Correct answer: (D)

Each box DOUBLES the dots: 1, 2, 4, 8 → next is **16**.

Q12. Correct answer: (B)

Yesterday's tomorrow IS today. So today is **Wednesday**.

Q13. Correct answer: (B)

First letters go forward (A, B, C, D); second letters go backward (Z, Y, X, W). Next: **DW**.

Q14. Correct answer: (B)

Total = $8 + 8 - 1 = 15$ (Veda is counted in both directions, so subtract 1).

**Q15. Correct answer: (A)**

South-East $\rightarrow$ North is a turn of 135° anticlockwise. East turned 135° anticlockwise becomes **North-West**.

Q16. Correct answer: (C)

16, 25, 36 and 49 are square numbers (4^2 , 5^2 , 6^2 , 7^2). **48** is not a square.

Q17. Correct answer: (B)

A mirror flips left-right. Mirror time = $12:00 - 3:00 =$ **9:00**.

Q18. Correct answer: (B)

Aman now = $2 \times 6 = 12$. After 3 years: $12 + 3 =$ **15**.

Q19. Correct answer: (C)

The rule is $\times 4$ then $+1$: $4 \rightarrow 17 \checkmark$ ($16+1$), $6 \rightarrow 25 \checkmark$ ($24+1$), so $8 \rightarrow 32 + 1 =$ **33**.

Q20. Correct answer: (B)

Order: Red $>$ Blue $>$ Green. So **Red is the heaviest**.

